Hot Chips Tutorial, Part-II: Evolution of enterprise open-source on RISC-V and architectural convergence on RVA23

Gordan Markuš
Director Silicon Alliances

$whoami

Gordan Markuš

Canonical, the company behind Ubuntu

- Electrical Engineer
- Open Source, Linux, Ubuntu
- Bridging Hardware and Code
- Football (soccer), skiing, photography

Instruction Set Architectures (ISA) support in Ubuntu

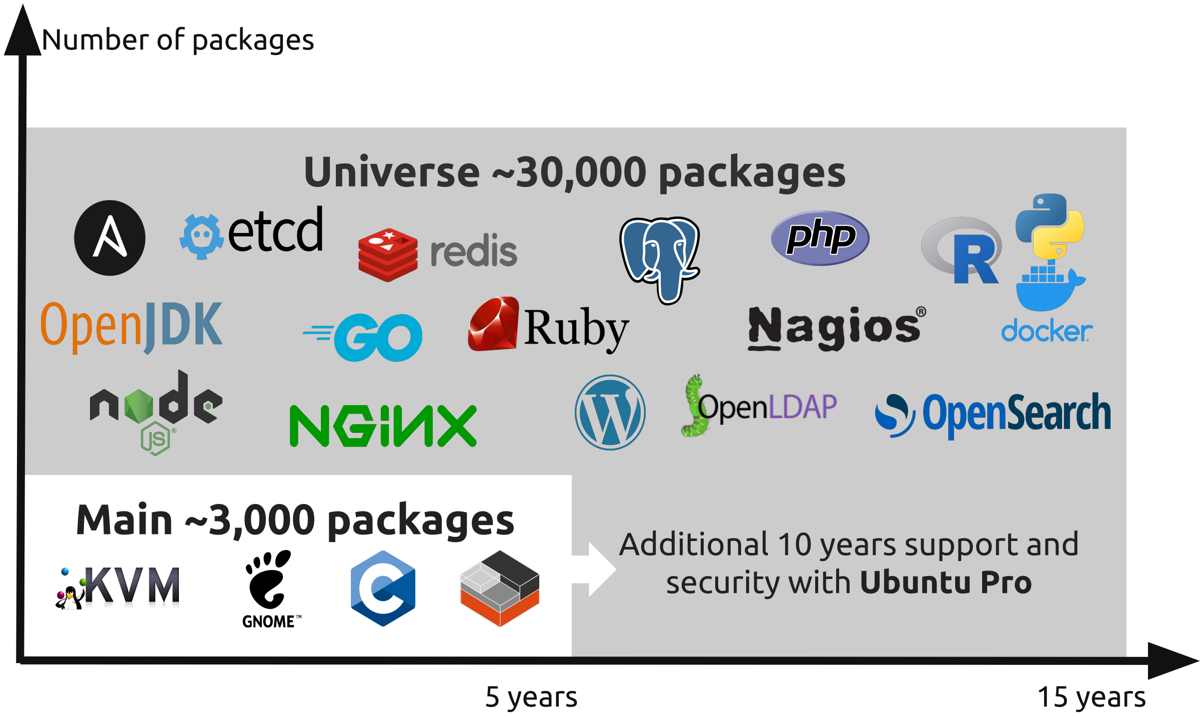
Building Ubuntu

Ubuntu Distribution = Kernel + User Space + Long-term Stability and Security

Ubuntu Universe Package Archive

Ubuntu Main Package Archive

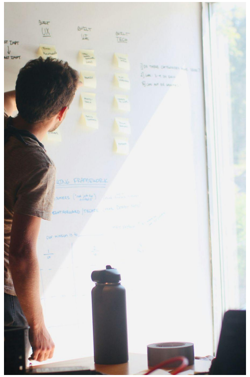
Ubuntu Kernel*



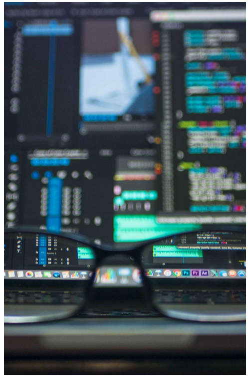
* Canonical has a portfolio of kernels optimized for our hyperscaler, silicon and other partners (https://ubuntu.com/kernel/variants)

Building Ubuntu

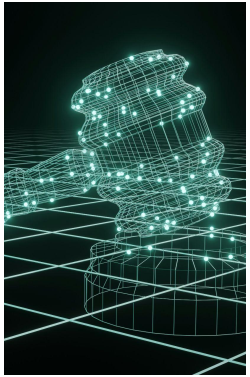
Release Planning Cycle



Merging and Feature Development



Stabilization and Freezes



Finalization and Maintenance

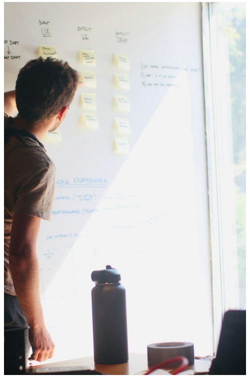
Final Release

Unified Ubuntu experience

Building Ubuntu

Our work is not done until we can long-term patch and maintain the released binary aggregation

Release Planning Cycle



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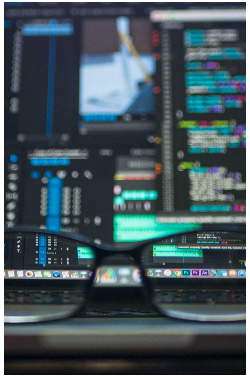

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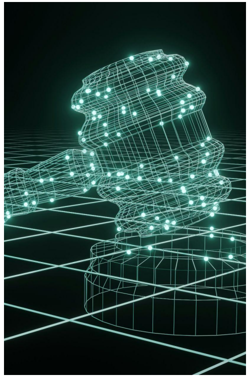


Finalization and Maintenance

Final Release

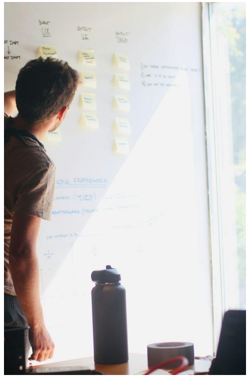
Unified Ubuntu experience

Building Ubuntu

Our work is not done until we can long-term patch and maintain the released binary aggregation

A Ubuntu system should always behave like a Ubuntu system, regardless of the ISA or platform

Release Planning Cycle



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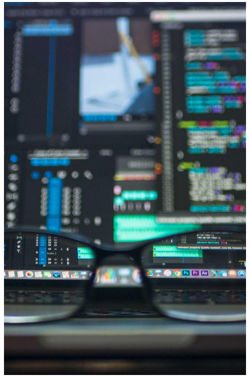

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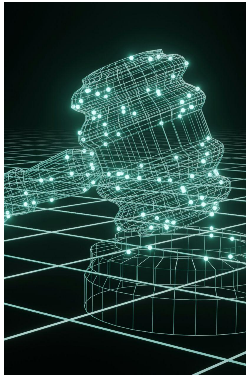


Finalization and Maintenance

Final Release

Unified Ubuntu experience

Ubuntu

Instruction Set Architectures (ISA) support in Ubuntu

ISA support in Ubuntu

Architecture support

Packages are typically built for each of several architectures. These are officially supported and maintained by the Ubuntu project.

Canonical provides server resources to build, store and distribute packages and installation media for them, and the core development team is responsible for their upkeep.

Architecture in this context is the ISA version.

Architecture baselines

For each architecture, there is a baseline which is the oldest or least capable ISA and extension set that can be used to run Ubuntu of that architecture.

ISA support in Ubuntu

Architecture	Baseline
`amd64`	Original 64-bit AMD / Intel CPUs (no mandatory extensions beyond baseline).
`arm64`	ARMv8-A baseline with VFPv4 and NEON. No optional extensions or ARMv8.1-A required.
`armhf`	ARMv7 with VFPv3-D16 floating-point support (NEON is not guaranteed).
`ppc64el`	• **Ubuntu 16.04 LTS – 21.10:** POWER8 or newer • **Ubuntu 22.04 LTS and beyond:** POWER9 or newer
`s390x`	• **Ubuntu 16.04 LTS – 19.10:** zEC12 or newer • **Ubuntu 20.04 LTS – 25.10:** z13 or newer • **Ubuntu 26.04 LTS and beyond:** z15 or newer
`riscv64`	• **Ubuntu 20.04 LTS – 25.04:** RVA20 profile • ***Ubuntu 25.10 and beyond: RVA23 profile***

https://ubuntu.com/project/docs/how-ubuntu-is-made/concepts/supported-architectures/

ISA support in Ubuntu

Architecture variants

To be able to assume features of more modern CPUs without dropping support for older machines, Ubuntu 25.10 introduced the concept of Architecture variant.

This allows packages to be built for given Architecture (in this context, architecture really means ABI) multiple times, each build assuming different CPU features.

The amd64v3 variant example

Ubuntu supports the **amd64v3** variant, which assumes the **x86-64-v3** microarchitecture level.

This assumes a number of instructions that have been added in the years since the first AMD64 CPUs, including AVX2, FMA, BMI2, and others.

Canonical's key objectives

Leading with early adoption

As an early adopter of new ISAs and emerging platform standards, Canonical provides the software foundation that others build upon.

Open source projects rely on availability of Ubuntu in order to start porting work on new platforms.

Canonical introduced arm64 support in 2011.

Maximum performance, zero compromises

Our ultimate mission is to make sure your hardware performs at its absolute best on Ubuntu and with Canonical's software stack.

We ensure full support for advanced features and platform-specific acceleration without compromising our security and support SLAs.

Not just CPUs but xPUs as well.

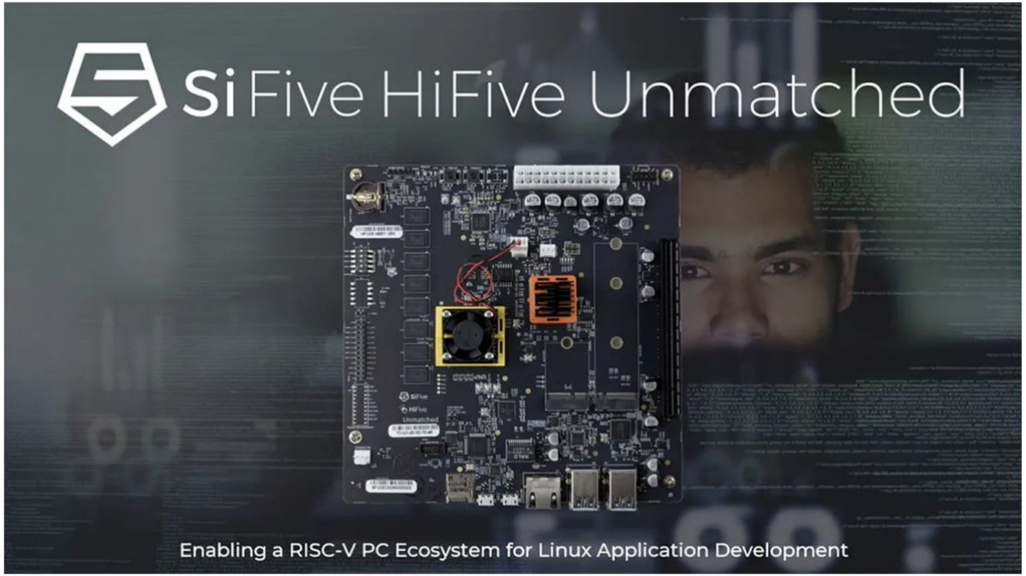
RISC-V and Ubuntu

The journey begins

Canonical Gives RISC-V a HiFive

News By Nathaniel Mott | Published June 24, 2021

Is Ubuntu to become the reference OS for RISC-V?



(Image credit: SiFive)

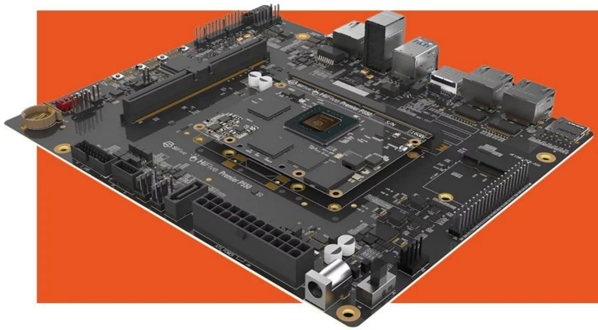
The journey continues

Canonical | Rivos

Announcement

Rivos and Canonical partner to deliver scalable RISC-V solutions in Data Centers

Learn more



SiFive

ESWIN

Canonical Ubuntu

XUANTIE | Canonical Ubuntu

Alibaba Damo Academy and Canonical partner to deliver Ubuntu on XuanTie and drive RISC-V innovation

Learn more

Canonical | SPACEMIT 进迭时空

Canonical

Blog
RISC-V

RISC-V: A 2025 retro and looking forward to 2026

Find out how Canonical's enablement efforts are driving RISC-V adoption

RISC-V®

RISC-V baseline

RVA23

64-Bit Target Profile

Standardized Application Baseline

RVA23 provides a milestone specification for 64-bit application processors. By requiring vector operations, hypervisor support, and bit manipulation out of the box, it delivers binary portability across diverse silicon while removing software porting ambiguity.

Evolving ISA standards and profiles

Linux Distribution Requirements (let's repeat)

- Stability and predictability
- Profiles simplify the RISC-V ecosystem by making sure hardware implementers and software developers roadmaps can intersect
- Canonical build systems requires upgrades and cross-team coordination to unlock a new major profile, this work is non-trivial

Status

- RVA23 profile ratification – **DONE**
- Server platform ratification – **DONE**
- Hardware availability and wide adoption – **server-class RVA23 hardware availability in 2026 and 2027 from multiple vendors**

Architectural variability exists across all major ISAs; RISC-V manages it through formal ratification processes and predictable application profiles.

Packages status and statistics

Archive Packages (debs)

~95%

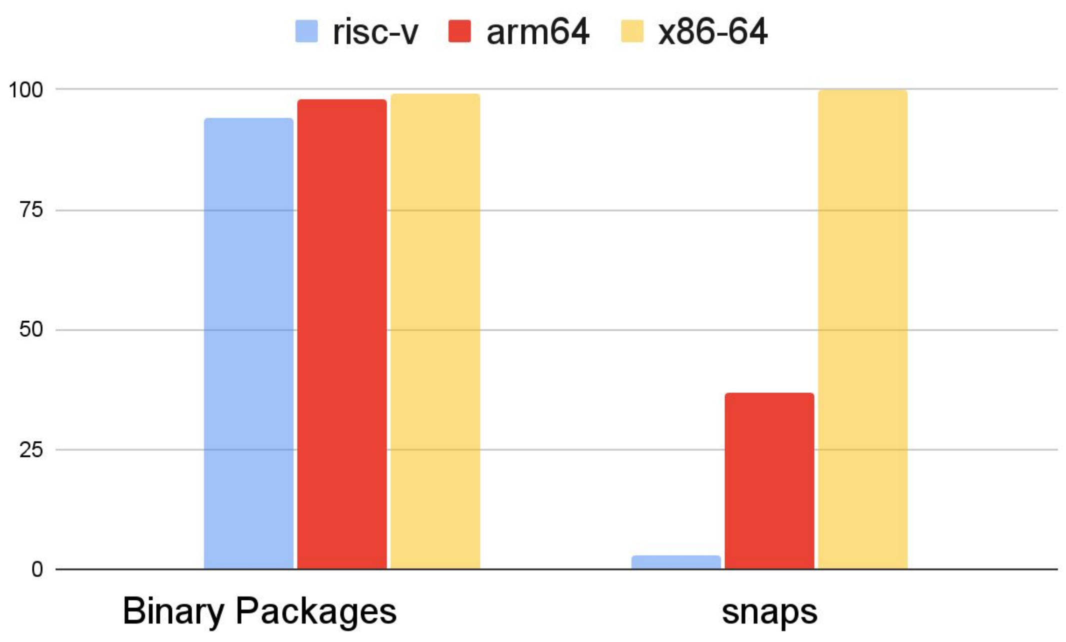
of packages are currently available on RISC-V compared to amd64.

Snaps Support

Work in Progress

Active development is underway to bridge the gap in snaps availability on RISC-V.

The majority of publicly available Snap packages target the client/desktop ecosystem.



Package Test Statistics Across Architectures

Architecture	amd64	amd64v3	arm64	armhf	ppc64el	riscv64	s390x
# Packages with Tests	24,815	20,763	24,770	24,559	24,735	**3,443**	24,533
# Packages Passing	22,900	19,869	22,908	21,404	22,878	**1,898**	21,486
Pass Rate (%)	**92.3%**	**95.7%**	**92.5%**	**87.2%**	**92.5%**	**55.1%**	**87.6%**

Release
Ubuntu 26.10 ("Stonking Stingray") is currently under test, this is a test release snapshot.

Commentary
While RISC-V package test coverage has significantly improved, a performance gap remains.
Emulated build and test targets have extended execution times from hours to days due to timeout issues and limited test volume.

Solution (already WiP)
Native RISC-V builders as a part of our build system infrastructure will significantly reduce build and test time of RISC-V packages.

Working closely with key partners to supply server-grade RVA23 platforms to Canonical.

Product Maturity Roadmap on RISC-V

Canonical Kubernetes

Porting Kubernetes and Canonical Kubernetes to provide resilient, lightweight, and scalable container orchestration natively on RISC-V nodes.

MicroCloud Integration

Automated, simple cloud deployment combining compute, storage, and networking into a unified zero-friction edge cloud solution.

Ceph and OVN Infrastructure

Native enablement of Ceph distributed object/block storage alongside OVN software-defined networking for enterprise reliability.

Private Cloud Infrastructure

Bringing Canonical's complete enterprise OpenStack and MAAS enabled private cloud stack to RISC-V, delivering full-featured multi-tenant cloud automation.

Dependency fixing and collaboration with key open source communities (next slide).

Open source collaboration

The **RISE Project** is focused on commercial software readiness in close partnership with RISC-V International to expedite delivery of more innovative RISC-V products into the market. The target market segments include mobile, consumer electronics, datacenter, and automotive.

Mission:

- Accelerate the development of open source software for RISC-V
- Raise the quality of RISC-V Platform software implementations
- Push the RISC-V Software ecosystem forward and align ecosystem partners' efforts

Open source collaboration

RISE
RISC-V Software Ecosystem

GitHub Runners

Easy hardware access for any GitHub project

- Install the **GitHub App**
- Native RISC-V runners on **Ubuntu 24.04**
- **24,000 jobs, 700k minutes, 116 organizations**
- Supports Llama.cpp, PyTorch, Numpy, k0s, and CNCF

Build & Board Farms

Build Farm

- Integration to Kernel, GCC, LLVM CI

Board Farm

- Partnering with RVI Lab Ecosystem WG
- Hosted at Scaleway Elastic Metal
- Supported by Alibaba and SpacemiT hardware

RISE Funded Upstream Software Enablement and Optimizations

LLVM • QEMU • Go • Rust • Python • OpenOCD • IREE • Llama.cpp • PyTorch

Conclusion

01
Standardized Governance

Profiles like RVA23 and the Server Platform turn architectural flexibility into predictable software targets. RISC-V and other ISAs have comparable working models.

02
Multiarch is the Standard

Mature multi-architecture paradigms lower the complexity of bringing software to new ISAs.

03
Ecosystem Momentum

Global upstream collaboration is closing software gaps and driving enterprise-grade maturity in RISC-V.

04
Canonical Leadership

Canonical leading the pace by delivering the software today, enabling solution build-out and collaboration ahead of broad hardware rollouts.

Thanks!